

Adam Labiosa

Curriculum Vitae

Madison, WI — labiosa@wisc.edu — adamlabiosa.github.io — linkedin.com/in/adam-labiosa

EDUCATION

Ph.D. Computer Science

09/22 – 05/28

University of Wisconsin-Madison
Department of Computer Science

- Advisor: Josiah Hanna
- Research focus: Reinforcement learning, robotics, meta-reasoning
- NSF NRT Fellow

B.S. Electrical Engineering and Computer Science

09/19 – 05/22

University of Wisconsin-Madison
College of Engineering

- Overall GPA: 3.94
- Graduated with Distinction

EXPERIENCE

Graduate Research Assistant

05/22 – 05/28

University of Wisconsin-Madison
Madison, WI

Advisor: Josiah Hanna

Current Research:

- Currently researching the environment characteristics that cause compute-constrained agents to meta-reason.
- Built an RL method for agents to meta-reason and act while planning occurs in real-time, dynamic domains, evaluated in Waymax and custom JAX environments (ICLR 2027 submission).
- Developed a learned meta-controller that selects between reactive control and deliberative planning and improved time-to-goal under latency costs (RLC 2026).
- Trained RL policies for multi-agent coordination and deployed on humanoid robots and achieved 3rd place at the RoboCup 2025 Champions Cup and 1st place in the RoboCup 2024 Challenge Shield (ICRA 2025).
- Leading a team of 20 masters and undergrad students in creating the lab's RoboCup system.
- Developing autonomous VLA-based systems for workplace robotics as part of an NSF NRT INTEGRATE fellowship.

Applied Science Intern

09/25 – 12/25

Amazon
New York City, NY

Research:

- Built a transformer-based offline RL model for multi-touch attribution, trained on over 1 million user interaction sequences.
- Evaluated against production baselines with improvement in attribution metrics.

Undergraduate Research Assistant

11/20 – 05/22

University of Wisconsin-Madison
Madison, WI

Advisor: Karu Sankaralingam

Research:

- Designed custom hardware accelerators for machine learning

Software Developer Intern

05/22 – 09/22

Blue Cross Blue Shield
Eagan, MN

Software Developer Intern

05/21 – 09/21

Blue Cross Blue Shield
Eagan, MN

PUBLICATIONS AND PREPRINTS

Labiosa, A., Orenstein, A., Bowling M., Hanna, J. (In preparation. Submission: JAIR October 2026) *Adaptive Computation in AI: A Survey*

Labiosa, A. & Hanna, J. (Submitted to ICLR 2027. Presented at the RL in Big Worlds Workshop at RLC 2026) *Learning to Act While Thinking*

Labiosa, A. & Hanna, J. (RLC 2026). *When to Plan: Learning to Select Between Reactive Control and Deliberative Planning*

Labiosa, A., Kimmel, Y., Korkut, M. (Submitted to Amazon Machine Learning Conference (AMLC) 2026). *Return Estimation with Transformers for Multi-Touch Attribution*

Labiosa, A.*, Wang, Z*, Agarwal, S., Cong, W., Hemkumar, G., Harish, A. N., Hong, B., Kelle, J., Li, C., Li, Y., Shao, Z., Stone, P.[†], & Hanna, J.[†]. (ICRA 2025). *Reinforcement Learning Within the Classical Robotics Stack: A Case Study in Robot Soccer*

- Award: Best RoboCup Themed Paper at the RoboLetics2.0 ICRA Workshop

Labiosa, A. & Hanna, J. (ICRA 2025). *Multi-Robot Collaboration through Reinforcement Learning and Abstract Simulation*

Corrado, N. Qu, Y., Balis, J. U., **Labiosa, A.**, & Hanna, J. (RLC 2024). *Guided Data Augmentation for Offline Reinforcement Learning and Imitation Learning*

AWARDS AND HONORS

NSF NRT Fellowship 2025

- Provides two years of funding for research.

3rd Place RoboCup Standard Platform League Champions Cup 2025

1st Place RoboCup Standard Platform League Challenge Shield 2024

University of Wisconsin-Madison Computer Science Research Scholarship 2023

3rd Place RoboCup Standard Platform League Challenge Shield 2023

Claude and Dora Richardson Engineering Scholarship 2022

Koch Family Foundation Scholarship 2019

TEACHING EXPERIENCE

Advanced Machine Learning - Teaching Assistant University of Wisconsin-Madison

- Lectured on reinforcement learning and AI ethics

Introduction to Artificial Intelligence - Teaching Assistant University of Wisconsin-Madison

Introduction to Computer Architecture - Teaching Assistant University of Wisconsin-Madison

Introduction to Computer Engineering - Teaching Assistant University of Wisconsin-Madison

SERVICE

Organizing Committee: RoboCup Humanoid Soccer League 2026.

Reviewer: NeurIPS (2024–2026), ICLR (2025–2026, Notable Reviewer both years), RLC (2025–2026), ICRA (2024–2025), AISTATS (2025–2026), IROS 2026, CogMI 2025, NLDL 2026, RoboCup Symposium (2024–2026).

SKILLS

Reinforcement learning, Robotics, Python, C++, Pytorch, Jax, MuJoCo, MJX, Brax, Ros2, Hydra, Docker, HTCondor HPC